Ernesto Contreras Alarcon

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Entry-level AI / Software Engineer (BSc First Class Honours) with hands-on experience building Python LLM systems: OpenAI Agents SDK, Amazon Bedrock, FastAPI, PostgreSQL and hybrid RAG. Built an academic advisor piloted with around 20 real users at CEU, and built a diagnostic agent with deterministic retrieval, fallback and pytest coverage. Use Claude Code and Codex daily. Available immediately and open to relocating.

PROFESSIONAL EXPERIENCE

CEU – IT Innovation Department Feb. 2025 – Jun. 2025
AI Software Engineer Intern Valencia, Spain

- Built an AI Academic Advisor with the OpenAI API and Agents SDK over institutional Snowflake data, returning structured JSON output; tested by around 20 students, tutors and staff.
- Wrote Python queries against Snowflake to extract the data and pass it into the agent's context, and managed its conversational memory.
- Found that the assignment due-date field was missing and that queries ran against Blackboard's generic schema, which led the team to CEU's own Snowflake copy.
- Iterated XML-structured prompts using user feedback: more relevant answers, fewer hallucinations and lower token consumption.

FEATURED AI PROJECTS

Enterprise Hybrid RAG Engine API | *Python, FastAPI, PostgreSQL, pgvector, Docker* 2026

- Built a hybrid retrieval API combining semantic vector search and lexical search using PostgreSQL, pgvector, HNSW, and GIN indexes.
- Implemented Reciprocal Rank Fusion (RRF) for document ranking and a PDF ingestion pipeline with recursive chunking (with overlap) and batched embedding generation.
- Added per-document isolation, multi-turn conversational memory and generated answers with page-level source citations, plus a web dashboard to upload and query PDFs.

Ring TelemetryOps – AI-assisted Diagnostic API | *Python, FastAPI, OpenCV, Amazon Bedrock, pytest* 2026

- Built a FastAPI diagnostic agent for Ring video doorbells for the Amazon Developer Hackathon 2026 (Ring track): it analyses video quality with OpenCV (frame rate, bitrate, sharpness, banding) and correlates it with simulated device and network telemetry to identify likely faults.
- Designed reproducible fault scenarios and a deterministic retrieval pipeline that ranks troubleshooting procedures from a 15-article knowledge base, with a local fallback when the model is unavailable.
- Integrated Amazon Bedrock (Claude) for reasoning; delivered REST endpoints, a CLI, an interactive web dashboard, and pytest suites that run without credentials.

WakeMap – AI Location & Travel Platform | *Flutter, Node.js, Express, Gemini API* 2025–2026

- Developed a cross-platform Flutter prototype with location-aware alerts and OpenStreetMap integration for journey planning.
- Built a Node.js/Express backend proxy that keeps the Gemini API key server-side and generates travel guidance from user context, time, budget, preferences, and walking-distance constraints.

EDUCATION

York St John University 2025 – 2026
BSc (Hons) Software Engineering – First Class Honours (74.50%) York, United Kingdom

Universidad CEU Cardenal Herrera 2023 – 2025
Higher Technician in Multiplatform Application Development (DAM) – Grade: 8.5/10 Valencia, Spain

Hugging Face 2026
AI Agents Course (LangGraph, smolagents, LlamaIndex, MCP, Langfuse) Online

TECHNICAL SKILLS

Backend & Data: Python, FastAPI, REST APIs, PostgreSQL, SQL, Snowflake, Node.js/Express, batch processing.
AI & LLMs: LLM APIs (Amazon Bedrock with Claude, OpenAI, Gemini), agents with tool calling and structured JSON outputs, hybrid RAG, embeddings, vector search (pgvector), XML-structured prompts, manual evaluation of response quality and token usage; OpenAI Agents SDK, LangGraph, smolagents, LlamaIndex, MCP, Langfuse (Hugging Face AI Agents course).
Programming: Python, SQL, TypeScript/JavaScript, Java; Go (basic).
AI-assisted Development: Daily use of Claude Code, Codex and Antigravity across development and testing.
Tools & Cloud: Git (command line), pytest, Docker, Azure (Azure AI Foundry), AWS (Amazon Bedrock), Firebase, n8n.
Languages: Spanish (Native), English (C1, UK university graduate).